Meshwa Hirpara

+91 97238 57964 | meshwahirpara23@gmail.com | LinkedIn | GitHub

Education

Vellore Institute of Technology — Bhopal, Madhya Pradesh
B.Tech. in Computer Science and Engineering – CGPA: 8.71/10.0 — *Oct. 2023 – May 2027*

Experience

Artificial Intelligence and Machine Learning Intern — May 2026 – Aug 2026
ParshWebCraft — *Udaipur, Rajasthan*

- Developed production-ready AI agentic applications using Python, MongoDB, and OpenRouter for AI-powered
- Built and deployed RESTful APIs for AI-powered automation and intelligent workflow execution.
- Improved backend performance and application reliability by implementing scalable software components.

Summer AI & Machine Learning Internship — Jun 2026 – Sep 2026
Technology Innovation Hub (TIH),IIT Guwahati — *Remote*

- Developed a YOLOv8-based object detection model for marine species detection.
- Trained and fine-tuned the model using Python, PyTorch, OpenCV, and Ultralytics YOLOv8.
- Performed data preprocessing,model training, and performance evaluation on computer vision datasets

Projects

Medical Insurance Cost Prediction [Link] | *Python, Scikit-learn, TensorFlow, SHAP, Streamlit* — Oct 2025

- Developed an insurance cost prediction model with 98.5% R^2 on 1,338 records using ML and Explainable AI
- Trained and evaluated 6+ ML/DL models, including Random Forest, CNN, and CNN-LSTM.
- Reduced prediction error by 40% compared to the baseline Linear Regression model
- Applied feature importance and SHAP-based Explainable AI techniques to interpret key factors influencing insurance.

YONO – Drone Vision Analytics [Link] | *Python, YOLO, OpenCV, Streamlit* — Aug 2026

- Developed a YOLO-based drone object detection system for automated analysis of aerial imagery, delivering real-time detection and visualization.
- Integrated YOLO, OpenCV, NumPy, Pandas, and Plotly to process, visualize,and analyze aerial image detection.
- Designed the system for 4+ real-world applications, including security surveillance, traffic monitoring, infrastructure inspection, and smart-city monitoring.
- Built an interactive Streamlit dashboard with configurable confidence thresholds, object counting,and detection.

Grain Price Prediction Analytics Dashboard | *Python,Chart.js,HTML, CSS* — Jan 2026

- Developed a web dashboard to analyze and visualize grain market prices using interactive charts.
- Integrated agricultural market data with commodity-wise and date-based filtering.
- Implemented automated price alerts based on configurable price thresholds.
- Built a responsive Flask application using Pandas and Chart.js for real-time market insights.

Technical Skills

Languages: Java, Python, Postgresql, HTML/CSS
Frameworks: FastAPI,Node.js, TensorFlow, Keras, PyTorch, Scikit-learn,Pandas, NumPy, Matplotlib, Streamlits
Developer Tools:Git, GitHub, VS Code, Google Colab, Jupyter Notebook,Microsoft excel
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Software Engineering
Analytics Skills: Data Analysis, Data Preparation, Data Visualization, Data Interpretation, Data Quality Validation, Power BI

Certifications

Machine Learning A-Z: AI, Python + ChatGPT Prize [2025]
Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate